

Limits and Derivatives

Section 1: Multiple Choice Questions

No calculator is allowed for these questions.

1. What is the value of $\lim_{h \rightarrow 0} \frac{\ln(e+h)-1}{h}$?
 - (A) 0
 - (B) $\frac{1}{e}$
 - (C) 1
 - (D) The limit does not exist.
2. Let $f(x) = 4x^3 - 2e^x + 5 \cos x$. What is the value of $f'(0)$?
 - (A) -2
 - (B) -7
 - (C) 3
 - (D) 5
3. What is the slope of the normal line to the graph of $y = x^2 - 4x$ at the point where $x = 3$?
 - (A) -2
 - (B) $-\frac{1}{2}$
 - (C) $\frac{1}{2}$
 - (D) 2
4. Which of the following limits represents the derivative of $f(x) = \sqrt{x}$ at $x = 9$?
 - (A) $\lim_{h \rightarrow 0} \frac{\sqrt{9+h}-3}{h}$
 - (B) $\lim_{x \rightarrow 9} \frac{\sqrt{x}-9}{x-3}$
 - (C) $\lim_{h \rightarrow 0} \frac{\sqrt{x+h}-\sqrt{x}}{h}$
 - (D) $\lim_{x \rightarrow 3} \frac{\sqrt{x}-3}{x-9}$
5. The graph of a function f has a sharp corner (cusp) at $x = 2$. Which of the following statements MUST be true?
 - (A) $f(x)$ is not continuous at $x = 2$.

(B) $\lim_{x \rightarrow 2} f(x)$ does not exist.

(C) $f'(2)$ does not exist.

(D) The tangent line to the graph of f at $x = 2$ is horizontal.

6. What is the value of $\lim_{x \rightarrow \frac{\pi}{4}} \frac{\sin x - \frac{\sqrt{2}}{2}}{x - \frac{\pi}{4}}$?

(A) 0

(B) $\frac{1}{2}$

(C) $\frac{\sqrt{2}}{2}$

(D) $-\frac{\sqrt{2}}{2}$

7. Let $g(x) = \frac{3}{x^2} - 4\sqrt[3]{x}$. Which of the following is $g'(x)$?

(A) $-\frac{6}{x^3} - \frac{4}{3}x^{-2/3}$

(B) $\frac{3}{2x} - \frac{4}{3}x^{-2/3}$

(C) $-\frac{6}{x^3} - \frac{4}{3}x^{2/3}$

(D) $-\frac{3}{x^3} - \frac{12}{x^{4/3}}$

8. Selected values of a differentiable function $v(t)$ are given in the table below.

t	1	3	5	7
$v(t)$	12	18	20	26

Which of the following is the best estimate for $v'(3)$ using the data in the table?

(A) 1

(B) 2

(C) 3

(D) 6

Section 2: Free Response Questions

Show all your work.

FRQ 1. Let f be the piecewise function defined by:

$$f(x) = \begin{cases} ax^2 + b, & \text{for } x \leq 2 \\ 5x - 1, & \text{for } x > 2 \end{cases}$$

where a and b are constants.

(a) If f is continuous at $x = 2$, write an equation relating a and b . Justify your answer using the definition of continuity.

(b) Find the values of a and b such that f is differentiable at $x = 2$. Show the algebraic work that leads to your answer.

(c) Using the values of a and b found in part (b), write the equation of the tangent line to the graph of f at $x = 1$.

FRQ 2. The temperature of a cup of coffee, in degrees Celsius, is modeled by a differentiable function $C(t)$ for $0 \leq t \leq 12$, where t is measured in minutes. Selected values of $C(t)$ are given in the table below.

t (minutes)	0	3	5	8	12
$C(t)$ ($^{\circ}\text{C}$)	90	75	60	52	45

(a) Use the data in the table to estimate the value of $C'(5)$. Show the computations that lead to your answer.

(b) Explain the meaning of $C'(5)$ in the context of the problem. Include correct units in your answer.

(c) Evaluate the expression $\lim_{h \rightarrow 0} \frac{C(3+h) - C(3)}{h}$. Briefly explain what this limit represents in the context of the problem.

FRQ 3. Consider the function g defined by $g(x) = 3e^x - 5x$.

(a) Find $g'(x)$ and evaluate $g'(0)$.

(b) Write an equation for the tangent line to the graph of g at $x = 0$.

(c) Find all values of x for which the tangent line to the graph of g is horizontal. Show the algebraic work that leads to your answer.